PROJECT INDRA — SYSTEMS ENGINEERING DOSSIER

Screen-2 Technical Portfolio: Blueprints, Schematics, Aerodynamic Derivations & Itemized BOM

Competition: SEWA FIRST Innovation Challenge 2026 (DTU ALTAIR & NEURAL AI)
Track: Disaster Management — Scout & Supervised-Delivery Drone (NAT-002-INDRA)

Project Lead: Dushant
Add-On Stack Cost: ₹5,600 (Budget: ₹5,300–₹6,000) | **Entry TRL:** 2

1. SYSTEM ARCHITECTURE & TWO-KEY SAFETY INTERLOCK

Project INDRA equips the competition's standard 10-inch quadcopter kit with an onboard autonomy layer for disaster scouting while strictly enforcing the non-negotiable **Manual Supervision Rule** through an independent two-key hardware and software interlock:

- **Key 1 (Target-Bound Cryptographic Token):** Casualties identified by the onboard 3-of-5 multi-frame voting pipeline (OpenCV ArUco and INT8-quantized TFLite) are queued by confidence on the GCS. When the operator validates the snapshot and selects *Confirm Drop*, a signed token bound to that target coordinate is generated with a strict 60-second expiration window.
- **Key 2 (Normally-Open Hardware Safety Relay):** Power to the MG90S release servo is physically interrupted by a Normally-Open (NO) relay controlled by companion computer GPIO 23. Only upon token verification and visual recheck at 3 m altitude does GPIO 23 energize the relay, allowing GPIO 18 to deliver the PWM release pulse. An avionics reboot or telemetry link loss leaves the servo completely unpowered.

2. OPTICAL DERIVATIONS & BALLISTICS SIMULATION

A. Ground Swath & Ground Sample Distance

At cruise search altitude $h = 10\ m$ with 62° HFOV and 1920 px sensor:

$$W = 2 \cdot h \cdot \tan(HFOV / 2) = 12.02\ m$$
$$GSD = 1202\ cm / 1920\ px = 0.63\ cm/px$$

Line spacing at 30% sidelap is 8.41 m. Sweeping a 50 × 50 m zone requires 6 passes (342 m) taking 85.5 s at 4 m/s.

B. Geotag Correction & Release Ballistics

Drop altitude $h_{drop} = 3.0\ m$; fall time & 2 m/s crosswind drift:

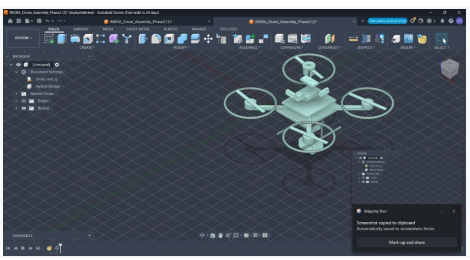
$$t_{fall} = \sqrt{(2h / g)} = \sqrt{(6.0 / 9.81)} = 0.782\ s$$
$$d_{drift} = v_{wind} \cdot t_{fall} = 2.0 \times 0.782 = 1.56\ m$$

Closed-loop visual centering over the ArUco marker directly eliminates standard 2–3 m GPS error, ensuring drop placement within < 1.5 m.

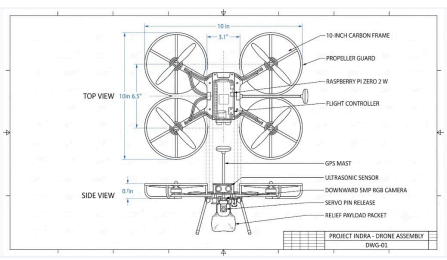
3. ITEMIZED BILL OF MATERIALS (ADD-ON ARCHITECTURE)

#	Subsystem Component	Engineering Specification	Qty	Cost (INR)
1	Companion Computer	Raspberry Pi Zero 2 W (1 GHz quad-core ARM, 512 MB RAM, Wi-Fi)	1	₹1,850
2	Vision Camera	5 MP OV5647 RGB Sensor (62° HFOV, 1080p, CSI ribbon interface)	1	₹750
3	Navigation & GNSS	u-blox NEO-M8N GNSS module + external compass on 92 mm mast	1	₹650
4	Obstacle Avoidance	2× HC-SR04 ultrasonic sensors (front and lateral) + divider resistors	1 set	₹280
5	Two-Key Release Unit	MG90S metal-gear servo, 5V NO optoisolated relay, 3D bracket	1 set	₹350
6	Power & Wiring Harness	5V 3A DC-DC buck converter, UART MAVLink cable, GPIO logic looms	1 set	₹420
7	Safety & Compliance	4× circular 3D-printed PETG propeller guards (R = 132 mm) + M3 hardware	1 set	₹450
8	Storage & Consumables	32 GB Class-10 microSD card, release latch pins, mock relief packet	1 lot	₹850
		TOTAL ADD-ON INVESTMENT (Airframe Kit Supplied by DTU):		**₹5,600**

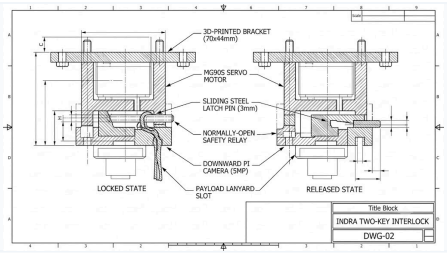
4. TECHNICAL DRAWINGS, CAD ASSEMBLIES & SCHEMATICS



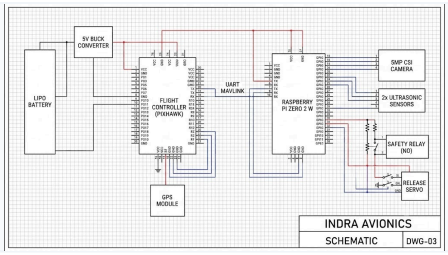
[PLATE 01] **3D Parametric CAD Assembly (Fusion 360)**



[PLATE 02] **Airframe Orthographic Projections (DWG-01)**



[PLATE 03] Two-Key Release Cross-Section (DWG-02)



[PLATE 04] Avionics & Power Schematic (DWG-03)